

56-100-650

Engine Test Certificate

Perkins Engine Company Limited
Eastfield, Peterborough PE1 5FQ
Great Britain

May 22, 2018



This engine has been tested to its maximum rating in accordance with Perkins defined requirements and has performed satisfactorily.

The electronics on this engine have been preset and when the engine has been run-in, the gross power output will be within +/-3% of the gross rating listed below, provided the exhaust and induction systems, fuel and lubricating oils are in accordance with Perkins defined requirements.

Engine Type	Perkins Serial Number	Customer
2806A-E18TAG2	JGBF5005N04723D	LPS2000

Reference	Technical Data
Emission Capability	Unregulated Low SFC
Engine Speed	50 Hz @1500RPM
Engine Type	2806A-E18TAG2
Fuel Consumption - Prime Power	202 g/kWh
Fuel Consumption - Standby Max	203 g/kWh
Gross Engine Power - Prime Power	584 kW
Gross Engine Power - Standby Max	628 kW
Gross Rating	584 kW
Net Rating	565 kW

This engine was built and tested in accordance with our approved ISO 9001 Quality Procedure and stated Rating, as determined by the ISO 3046 (BS5541) Standard.

Performance certified as satisfactory.

Signed,

For and on behalf of Perkins Engines Company Limited - Large Engines Division.

MUST BE KEPT WITH ENGINE AT ALL TIMES

Issue 4

Test Certificate

FOR

SALIENT POLE, SELF EXCITED / PMG EXCITED AND SELF REGULATED AC GENERATOR

Machine No. **N19G304574**

Type Test has been conducted on similar design machine
STANDARDS: Generator generally conforms to BS 5000 : Part 3,
 Tested in accordance with BS EN 60034-1/IS/IEC 60034-1
 Rotor dynamically balanced to BS ISO 1940 - 1

ROUTINE TEST CERTIFICATE FOR BRUSHLESS AC GENERATOR:

FRAME: HCI544F1	AVR: AS440	AVR NO: 191401480248	ENCLOSURE: IP23	
RATED KVA: 650	RATED VOLTS: 415	RATED AMPS: 904.3	FREQUENCY: 50	PHASE: 3
POWER FACTOR: .8	STR CONN: Series Star	STR WINDING: 311	DUTY: Base Continuous	WIRE: 4-Wire
EXC. VOLTS: 48	EXC.AMPS: 2.3	AMB TEMP: 40		

FOR 3 PHASE SEQUENCE LOOKING FROM DRIVE END FOR CLOCKWISE ROTATION :U-V-W

WINDING	COLD RESISTANCE IN Ohm(+/- 10 Milli Ohm) AT 20°C (MAX VALUES)	INSULATION RESISTANCE (M- Ohm)	HV TEST 2KV FOR 1 MIN	INSULATION CLASS
STATOR	0.0078	2+	OK	H
ROTOR	2.284	2+	OK	H

REGULATION TEST

FREQ.	VOLTS	AMPS	KVA	PF	KW	EXC.VOLT	EXC.AMP	% LOAD
52	416.4	0.00	0.00	0.00	0.00	10	.54	0.00
50	415	904.3	650	.8	520	48	2.3	100.00

REGULATION = (N.L.VOLTAGE-F.L.VOLTAGE / RATED VOLTAGE)*100 = 0.34 %

Item Description: AC Generator.HC5.4.F.1.311.,0,,18,.AS440.Black - RAL9011
 Remarks:

Date: 26-JUL-19	Job No: 3749706-003	AUTHORISED BY Shubham S Pawar Quality Assurance
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Certificate Of Conformity

Quality System Certification
 ISO 9001:2015

Frame Size : **HCI544F1**Serial Number : **N19G304574**

Manufactured Date : 26-JUL-19

KVA: **650**Authorized By Quality Assurance: **Shubham S Pawar**

EU DECLARATION OF CONFORMITY



**Generator
Technologies**

This synchronous A.C. generator is designed for incorporation into an electricity generating-set and fulfils all the relevant provisions of the following EU Directive(s) when installed in accordance with the installation instructions contained in the product documentation:

2014/35/EU	Low Voltage Directive
2014/30/EU	The Electromagnetic Compatibility (EMC) Directive
2011/65/EU	Restriction on Hazardous Substances in Electrical and Electronic Equipment (RoHS) Directive
2015/863	Delegated Directive amending Annex II of 2011/65/EU

and that the standards and/or technical specifications referenced below have been applied:

EN 61000-6-2:2005	Electromagnetic compatibility (EMC). Generic standards - Part 6-2: Immunity for industrial environments
EN 61000-6-4:2007+A1:2011	Electromagnetic compatibility (EMC). Generic standards - Part 6-4: Emission standard for industrial environments
EN ISO 12100:2010	Safety of machinery - General principles for design - Risk assessment and risk reduction
EN 60034-1:2010	Rotating electrical machines - Part 1: Rating and performance
BS ISO 8528-3:2005	Reciprocating internal combustion engine driven alternating current generating sets - Part 3: Alternating current generators for generating sets
BS 5000-3:2006	Rotating electrical machines of particular types or for particular applications - Part 3: Generators to be driven by reciprocating internal combustion engines - Requirements for resistance to vibration
EN 50581:2012	Technical documentation for the assessment of electrical and electronic products with respect to the restriction of hazardous substances

This declaration has been issued under the sole responsibility of the manufacturer. The object of this Declaration is in conformity with the relevant Union harmonization Legislation.

The name and address of authorised representative, authorised to compile the relevant technical documentation, is the Company Secretary, Cummins Generator Technologies Romania, B-dul Decebal Nr. 116A 200746 Craiova Dolj, Romania.

Signed:

Date:

6th March 2019

Name, Title and Address:

Kevan J Simon
Global Technical Director
Cummins Generator Technologies Romania
B-dul Decebal Nr. 116A
200746, Craiova
Dolj, ROMANIA

Description HCI544F1

Serial Number N19G304574

EU DECLARATION OF CONFORMITY



**Generator
Technologies**

The A.C. Generator utilizes hazardous material exemptions as detailed in Annex III of EU Directive 2011/65/EU

Products carrying the following descriptions are considered to be out of scope of RoHS Directive 2011/65/EU, intended to be installed in Large Scale Fixed Installations and for installation into a predefined and dedicated location, installed and de-installed by professionals:

LVI80*

LVSI80*

DSG 99*

DSG 114*

DSG 125*

DSG 144*

Where "*" represents any combination of letters and characters completing the specific description of the product.

450-16383-G